

AI-Powered Enterprise Meeting Intelligence and Action Tracking Platform

Dr K Anand
Professor, Computer Science and
Engineering
Rajalakshmi Engineering College
Chennai, India
anand.k@rajalakshmi.edu.in

Girija S P
Student
Computer Science and Engineering
Rajalakshmi Engineering College
Chennai, India
230701092@rajalakshmi.edu.in

Bhuvaneshwari K
Student
Computer Science and Engineering
Rajalakshmi Engineering College
Chennai, India
230701055@rajalakshmi.edu.in

Abstract - Many organizations hold meetings to discuss projects; assignments and progress on said projects are assigned and tracked. Such meetings often require taking notes and recording action items, which can be tedious and time-consuming, resulting in missed action items, vague instructions, and poor collaboration, for example, when meetings are held with colleagues on the go; with spotty internet access; or in which taking notes is otherwise difficult. The proposed AI-Powered Enterprise Meeting Intelligence and Action Tracking Platform would serve to address this need while enabling easy processing of action items, summarization of discussions, and tracking of these items. The proposed platform would be used for both online and offline meetings. It would take an audio file of the meeting and process it with Artificial Intelligence to identify and extract action items, noting who is responsible for them and any relevant deadlines. Information would be presented in an easy-to-use interface, with meeting history and search functions available, and users could be reminded of tasks. The proposed system would make it easy and intuitive for enterprises to increase employee productivity and collaboration, by transforming natural language discussions into actionable items that move projects forward, and ensuring that nothing is forgotten. The system would be organized around a web interface and would be used in a variety of collaboration settings.

Keywords: Artificial Intelligence, Meeting Intelligence, Meeting Summarization, Action Item Extraction, Enterprise Collaboration, Natural Language Processing, Task Tracking.

I. INTRODUCTION

Meetings are one of the essential elements of the communication process in any organization. People conduct meetings with their subordinates, superiors, peers, and customers to deliberate on project details, performance reviews, work distribution, issue resolution, and other business decisions. Even though meetings are critical, collecting and organizing the information resulting from such events is a significant challenge.

Typically, employees keep notes to record the meeting details, decisions, tasks, and deadlines. At the same time, taking notes is a hindrance when one is on the move, at noisy places, in online conferences with bad internet connection, and working on multiple tasks. Therefore, crucial information may

be missed or forgotten after the meeting because one might forget to note down the tasks to do after the meeting ends.

format to extract the actionable insights. The system will be designed to capture the recordings of online and offline meetings and summarize the transcripts to identify the key discussion areas, decisions made, action items, owners, and deadlines. It is the responsibility of the proposed system to guarantee that the information captured in a meeting is transformed into organization knowledge managed to monitor responsibility. The system will also provide a collaborative workspace where users access the results of previous meetings, search for information, and follow up on identified action items and their progress.

II. LITERATURE REVIEW

The paper of Janice D. Otter, John R. B. H. et al. studies automatic speech recognition for meeting conversations. One of the paper's main points is recognition of meeting content speech into the textual representation that can be later processed by different natural language processing means. Speech recognition, unlike manual meeting transcription, can be helpful in saving efforts on recording long meetings [1], [2]. However, recognition accuracy can be hampered by multiple factors, including background noise, overlapped speech, poor speaker accent, and audio quality issues [3]. The approach suggested in the paper mainly addresses the task of speech-to-text conversion, without attempting to extract any meeting decisions, assign actions, track deadlines, and follow-up items.

On the contrary, the paper of McCowan et al. (2005) [4] addresses the problem of meeting speech recognition relying on multiple information sources. The approach takes into account features of meeting conversations aiming to enhance recognition of spontaneous speech. The paper's results show that meeting speech has a number of distinct features, compared to a well-articulated conversation. The reason is that meeting participants often interrupt each other, speaking at the same time. While the approach allows for automatic transcription of meetings, this transcription still needs to be processed to extract relevant actionable outcomes. The approach suggested by the paper does not include a procedure that would automatically recognize task owners and track action items.

Zechner (2002) [5] explored automatic summarization of spoken dialogues and meeting-related conversations. A study was conducted to determine the critical information that could be extracted from extended segments of speech. Such summaries can potentially minimize the amount of data a person needs to process to obtain relevant information. However, the summarization techniques described by the author may not always be sufficient to clearly differentiate between a general discussion within the conversation and specific actions like decisions, assignments, and deadlines [6]. Thus, it may still be necessary for a person to review the transcript to identify what action items were decided during the meeting.

The work by Murray, Carenini, and Ng (2005) [7] also dealt with the automatic summarization of meetings but focused on multiparty conversations. It can be noted that the described approaches allow for extracting the key information from lengthy transcripts. This information typically includes actions that were decided during the meeting. However, as mentioned above, meeting summaries that include such actions are not sufficient for enterprise task management. This aspect is critical for many organizations since they typically require a detailed schedule of tasks along with their deadlines and responsible parties. Thus, it may be beneficial to combine the described automatic summarization techniques to manage enterprise tasks while also extracting the relevant information from meetings.

Finally, Janin et al. (2003) [8] introduced the ICSI Meeting Corpus, which has become an important source of data for the automatic processing of meetings. The corpus includes meetings as audio recordings and transcripts for further processing and analysis. This data can be used to build speech recognition, speaker recognition, dialogue act tagging, and summarization models, as described by the authors. The main benefit of this research is that it allows computational linguistics to be applied to real-world meetings. However, it is evident that the corpus is mainly a supporting tool for the aforementioned models. Thus, it is unclear how these models can be integrated into a single solution for enterprise meeting management, including automatic assignment of tasks, tracking of deadlines and progress, and other features.

The paper by Murray & Carenini (2008) [9] addresses the task of extractive summarization of meeting conversations based on the information extracted from dialogues. The method proposed allows automatically extracting the most informative parts of meetings for faster and more efficient processing. Such methods may be effectively used for minimizing the time spent on reviewing meetings. However, extractive approaches are typically not sufficient for enterprise-level applications since they do not fully decompose the input information. In particular, for meetings, in addition to the key ideas, there are usually decisions, action items, responsible employees, and deadlines that should be explicitly identified.

With the recent advances in the field of large language models, specifically those based on the transformer architecture [10], the task of meeting summarization can be effectively addressed [11], [12]. State-of-the-art generative models allow extracting and rephrasing the most important parts of the conversation in a much more efficient way. Moreover, the latest implementations address multi-turn dialogues, allowing capturing the context of a multi-paragraph conversation [13]. However, applying such models to hold meetings is not sufficient since such models are likely to invent some information. To address this problem, the model should get proper prompts, response templates, data validation, and instruction to extract or generate additional information only when it is explicitly mentioned in the conversation.

III. METHODOLOGY

The proposed AI-Powered Enterprise Meeting Intelligence and Action Tracking Platform is envisioned as a centralized web-based system that transforms meeting recordings into intelligible information. The system encompasses both online and offline meeting scenarios wherein a user can upload meeting recordings for processing when they are not able to take notes manually due to traveling, limited internet access, or other limitations. The system is partitioned into several modules that take care of the input of the meetings recording, management of the recorded data, artificial intelligence (AI) processing, information extraction, and action tracking. The overall design partitions the user-interface, backend processing, and AI services for ease of maintenance and extensibility in the future. The process commences with the recording of a meeting followed by the recording upload, AI processing, information extraction, and the ultimate presentation of the information in the meeting dashboard.

The first component of the system is the Meeting Input and Recording Module, which takes the input of the recorded meeting. The recorded meeting may either be an online or an offline recording. Once a user selects the recording they intend to upload, the system facilitates the upload of the recording to the backend server through the web application. The backend system then validates the uploaded recording and links it to the respective meeting. This approach allows employees to submit the recording of a meeting even when they are not in a position to take notes digitally or manually at the moment of the meeting.

The uploaded recording is then processed in the AI Processing Module, which utilizes the Gemini AI model [14] to comprehend the contents of the recorded meeting. The AI processes the uploaded recording to extract the relevant information contained within the body of the meeting. The process performed by the AI is geared toward the extraction of the relevant enterprise information as opposed to storing the actual recording of the meeting. The information extracted by the AI is structured according to the types of information

specified in the system and stored in the database as separate entities including summary, key discussion, decisions, action items, assignees, and deadlines.

A. MEETING INPUT AND RECORDING MODULE

The Meeting Input and Recording Module is the first component of the proposed system. The module is geared towards collecting the meeting records input from the user and uploading the data to the backend system for processing. The Meeting Input and Recording Module handles both online and offline meeting recordings.

In the case of an online meeting, the user can directly utilize an online communication tool to host a meeting and record the meeting. The recorded video can then be uploaded into the system by navigating into the system’s dashboard and selecting the upload option. On the other hand, in the case of an offline meeting, the user would need to utilize an audio or video recording tool to record the meeting and upload the recording as explained for online meetings. The file uploaded by the user is received by the web interface through a multipart file-upload procedure. The backend system facilitates the processing of the uploaded recording through a file-upload service that stores the recording temporarily before it is processed by the AI Processing Module.

B. AI PROCESSING MODULE

The AI Processing Module is the main body of the proposed system. Upon a successful recording upload, the backend will be tasked to relay the recording to the Google Gemini AI Service for processing. The AI model will analyze the conversation of the meeting recording to extract the relevant information contained within it [15], [16]. A structured format will be used to present the resulting information so that it can be easily presented by the front end.

The AI processing will result in the following major outputs:

- 1. Meeting summary
- 2. Key discussion points
- 3. Decision
- 4. Action items
- 5. Who to assign tasks to
- 6. Due dates
- 7. Important deadlines

The system will be programmed to only use the information contained within the meeting for its processing. The system will also be designed not to assume the presence of certain categories of information if it is not present in a recording to avoid the inclusion of unsubstantiated details.

C. MEETING INTELLIGENCE MODULE

The Meeting Intelligence Module is tasked with transforming the information generated by the AI processing into meaningful meeting insights. Instead of having to listen to

the entire recording of a meeting by the employees or the managers, the system will generate the relevant insights that were identified in a meeting. Summary component Key discussion points component Decision component. The summary component will provide a brief overview of the details of the meeting. The key discussion points component will highlight the major issues that were brought up by the participants. The decision component will indicate the decisions that were made during the meeting. These components will allow the employees to be able to follow the outcome of a meeting without having to listen to the entire recording. The information extracted from the meeting by the AI processing will be presented using the meeting results dashboard so that it can be easily reviewed.

D. ACTION ITEM AND TASK TRACKING MODULE

The Action Item and Task Tracking Module functions in relation to discussion results with attached tasks. In enterprise-level meetings, tasks can be assigned to employees and have to be tracked, although they may not always be properly documented. The proposed system would extract such action items from the meeting and organize them using the action task, assignee, and due date, if stated in the discussion [6]. The action task provides the objective that has to be completed, while the assignee is an employee who was given this assignment. The extracted action items would be divided into particular tasks and assigned to correspondents in order to track their progress. Such items extracted from the discussion would be separated from the summary in order for the user to distinguish between objectives that have to be completed and the current state of these objectives. The Action Item and Task Tracking Module would bridge the meeting and its aftermath, addressing the results and assigning tasks to process them.

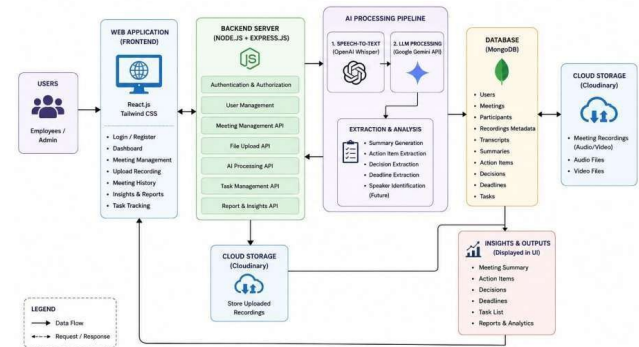


Figure 3.1. Architecture Diagram of the system

E. DEADLINE AND REMINDER MODULE

The Deadline and Reminder Module addresses the deadlines and reminders set for action items and discussion results. They should be organized based on the content of the meeting.

The deadlines extracted by the AI can be matched with the information in the meeting to provide context for the user. In such a way, the user would be able to track the pending items

and their deadlines without searching through the entire recording for them. With the addition of the reminder feature available in the enterprise plan, the user would be able to keep track of their tasks and the corresponding deadlines to follow.

This module would be especially useful in the enterprise plan due to multiple meetings scheduled with the same person, which would have multiple deadlines. Keeping track of the pertinent dates would decrease the chance of human error.

F. MEETING HISTORY MODULE

The Meeting History Module would be utilized to keep an archive of the processed meetings, which would allow retrieving the information from them if necessary. Instead of relying on personal memory, the user would be able to utilize the data from the meetings, which was extracted either as the summary or in another form, depending on the user’s preferences.

Every processed meeting would have to be stored in order to retrieve the summary or any other information from them. The Meeting History would offer a useful function of going through the past meetings to view the contents for later reference.

Furthermore, the Meeting History would allow additional enterprise-level features to be developed as updates for the users.

G. INTELLIGENT SEARCH MODULE

The intelligent search module will be designed to ensure effective search of the information related to the decisions made during the meetings. Typically, in a meeting management system, a user would have to scan through multiple meeting records or files to identify the required information.

In the proposed system, all information extracted from the meetings will be stored in a database that can be searched using specific terms or meeting-related data fields. Thus, finding the information related to the decisions made during the meetings will be automated and much faster. The search functionality can later be augmented with semantic and vector search engines to find related meeting data.

H. WEB APPLICATION AND BACKEND MODULE

The proposed system will be based on the client-server architecture. The front end will comprise the web application, whereas the back end will be hosted as a web service. The front end will be the interface through which the users will be able to upload the audio files, process the information, and view the results.

The back end will host the web application, respond to front-end requests, store recordings, and support the complete meeting lifecycle, including the interaction with the AI service. Essentially, the backend will act as an intermediary between the front end and the AI service. The web application front end will be developed in React and Vite framework,

whereas the backend will be written in Node.js and Express.js programming languages. The communication between the front- and backend will be established via REST API.

The backend will primarily host all the data, respond to front-end queries, process the recordings, and store them for the AI service to consume. Overall, the frontend and backend will be relatively straightforward and modular, allowing for easier maintenance and enterprise-level development in the future. assigns toxicity scores to all the given input data, with the help of the BERT encoder and classifier [17]. The model’s demographic attributes sometimes contribute to varying degrees of bias, which makes the final results misleading, hence bias elimination is executed in this package alongside the sentimental analysis.

IV. RESULT AND ANALYSIS

Implementation and Analysis of the Proposed AI-Powered Enterprise Meeting Intelligence and Action Tracking Platform Essay. This section presents the implementation and analysis of the proposed AI-Powered Enterprise Meeting Intelligence and Action Tracking Platform. The developed system offers a complete solution for uploading meeting recordings, processing them with an artificial intelligence model, and extracting critical information from the conversations held during meetings. The results demonstrate how the unstructured audio or video recording can be converted into a highly-structured meeting information resource.

The Figure 4.1 below represents the meeting recording upload interface. Users can enter an audio or video recording of an online or physical meeting. The Figure 4.2 below represents the AI-based processing stage of the proposed system. After uploading the recording, the backend transfers it to the AI processor. The Figure 4.3 below represents the AI-generated meeting summary. The summary provides a brief overview of the entire meeting, allowing the user to become aware of the general subject of the conversation without listening to or reading the entire recording.

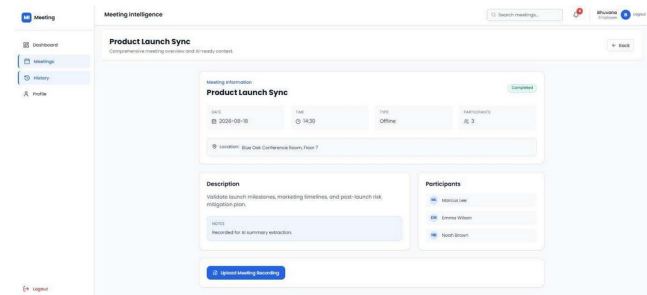


Figure 4.1 Meeting recording upload interface.

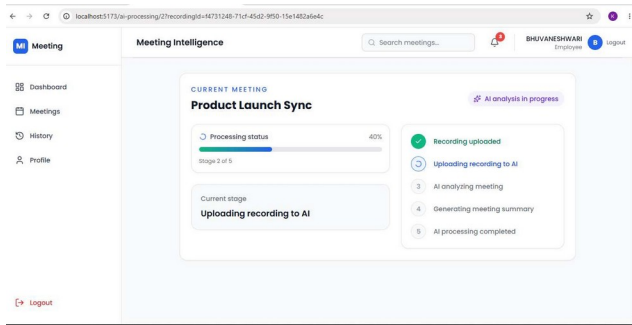


Figure 4.2 AI-based processing stage

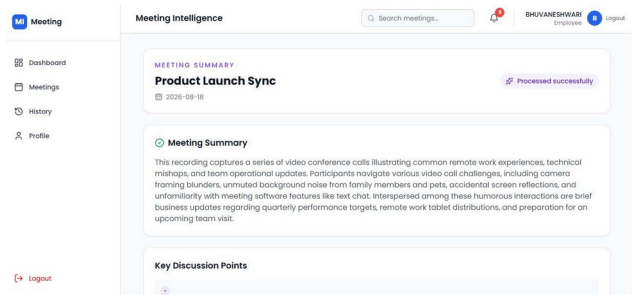


Figure 4.3 Meeting Summary

The Figure 4.4 below represents the key discussion points extracted from the meeting. The essential points contain the primary issues brought up during the meeting. In addition, separating the discussion points from the entire conversation allows the user to quickly review the main subjects of the conversation. The Figure 4.5 below represents the decisions extracted from the meeting. Similar to discussion points, decisions are separated from the conversation and displayed as a distinct section.

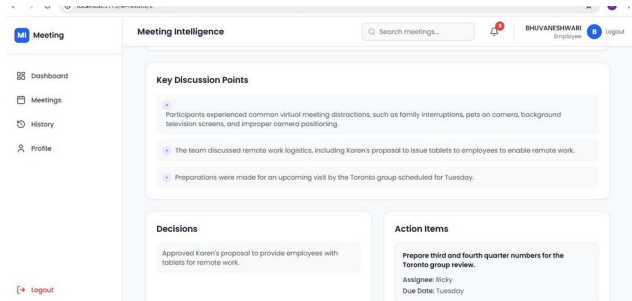


Figure 4.4 Key Discussion Points

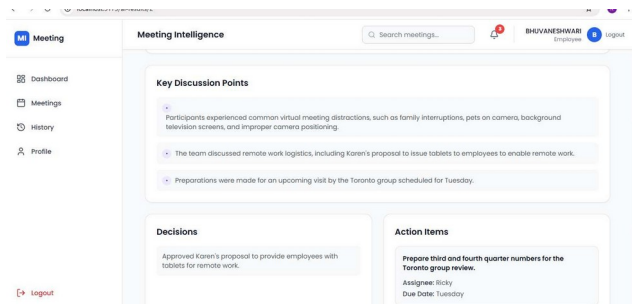


Figure 4.5 Decisions

The Figure 4.6 below represents the extracted action items with assignees and due dates. The system processes the

conversation and identifies action items – tasks that should be performed after the meeting. In addition, the AI model attempts to detect assignees – persons or teams responsible for performing particular action items. The Figure 4.7 below represents the extracted deadlines from the meeting. Similar to the action items, deadlines are separated from the conversation and displayed as a distinct section. This way, users obtain a clear view of the essential dates associated with the meeting.

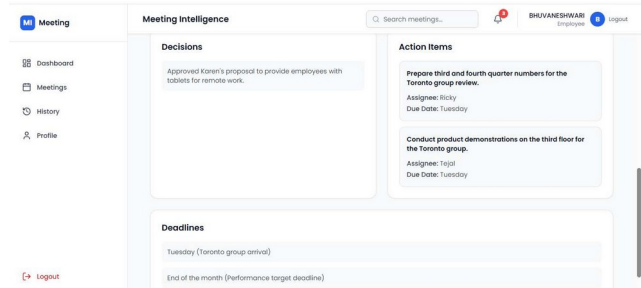


Figure 4.6 Assignees and Due dates

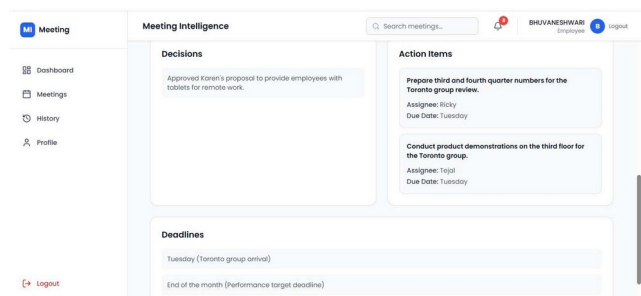


Figure 4.7 Deadlines

The Figure 4.8 below represents the meeting history interface. The system stores the information on all processed meetings, allowing the users to review the meetings' details if needed. This way, the users can track the processing of their recordings and re-access the meeting details at any time.

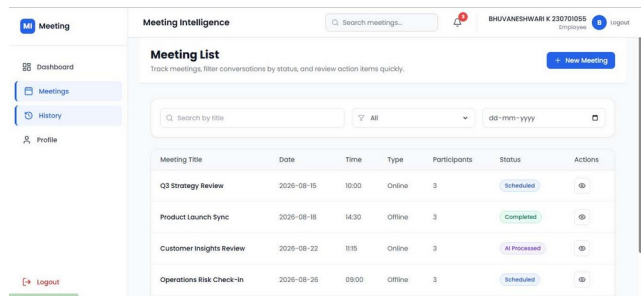


Figure 4.8 Meeting History

V. CONCLUSION

In this paper, we have discussed the challenges of traditional enterprise meeting management and proposed an AI-Powered Enterprise Meeting Intelligence and Action Tracking Platform to address these challenges. This proposed system offers a centralized web-based enterprise system for processing both online and offline meeting recordings into actionable information.

Our proposed system is powered by artificial intelligence to generate meeting summaries, key discussion points, decisions, action items, assignees, and deadlines from recorded meetings, reducing the manual effort of preparing minutes of meetings and helping users focus on the critical deliverables and outcomes of the meetings.

Therefore, our proposed system shows the potential of artificial intelligence to transform the traditional approach of managing meetings into an intelligent workflow, transforming the discussion into knowledge and decisions, identifying the action items, and linking tasks to the persons responsible with deadlines.

VI. REFERENCES

- [1] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust Speech Recognition via Large-Scale Weak Supervision," arXiv:2212.04356, 2022.
- [2] A. Baevski, H. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations," in *Advances in Neural Information Processing Systems* 33, 2020.
- [3] J. Carletta, S. Ashby, S. Bourban, M. Flynn, M. Guillemot, T. Hain, J. Kadlec, V. Karaiskos, W. Kraaij, M. Kronenthal, G. Lathoud, M. Lincoln, A. Lisowska, I. McCowan, W. Post, D. Reidsma, and P. Wellner, "The AMI Meeting Corpus: A Pre-announcement," in *Machine Learning for Multimodal Interaction, Second International Workshop, MLMI 2005, Revised Selected Papers, Lecture Notes in Computer Science*, vol. 3869, Springer, 2006, pp. 28–39, doi: 10.1007/11677482_3.
- [4] I. McCowan, D. Gatica-Perez, S. Bengio, G. Lathoud, M. Barnard, and D. Zhang, "Automatic analysis of multimodal group actions in meetings," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 27, no. 3, pp. 305–317, Mar. 2005.
- [5] K. Zechner, "Automatic summarization of open-domain multiparty dialogues in diverse genres," *Computational Linguistics*, vol. 28, no. 4, pp. 447–485, Dec. 2002, doi: 10.1162/089120102762671945.
- [6] Y.-N. Chen and D. Hakkani-Tür, "AIMU: Actionable Items for Meeting Understanding," in *Proceedings of the Tenth International Conference on Language Resources and Evaluation (LREC'16)*, 2016, pp. 739–743.
- [7] G. Murray, S. Renals, and J. Carletta, "Extractive summarization of meeting recordings," in *Proceedings of the 9th European Conference on Speech Communication and Technology (Interspeech 2005 – Eurospeech)*, Lisbon, Portugal, 2005, pp. 593–596.
- [8] A. Janin et al., "The ICSI Meeting Corpus," in *Proceedings of the 2003 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP '03)*, Hong Kong, 2003, doi: 10.1109/ICASSP.2003.1198793.
- [9] G. Murray and G. Carenini, "Summarizing Spoken and Written Conversations," in *Proceedings of the 2008 Conference on Empirical Methods in Natural Language Processing*, Honolulu, HI, USA, Oct. 2008, pp. 773–782.
- [10] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention Is All You Need," in *Advances in Neural Information Processing Systems* 30, 2017.
- [11] M. T. R. Laskar, X.-Y. Fu, C. Chen, and Shashi Bhushan TN, "Building Real-World Meeting Summarization Systems using Large Language Models: A Practical Perspective," in *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing: Industry Track*, 2023, pp. 343–352, doi: 10.18653/v1/2023.emnlp-industry.33.
- [12] F. Kirstein, T. Ruas, R. Kratel, and B. Gipp, "Tell me what I need to know: Exploring LLM-based (Personalized) Abstractive Multi-Source Meeting Summarization," in *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*, 2024, pp. 920–939, doi: 10.18653/v1/2024.emnlp-industry.69.
- [13] M. Zhong, D. Yin, T. Yu, A. Zaidi, M. Mutuma, R. Jha, A. H. Awadallah, A. Celikyilmaz, Y. Liu, X. Qiu, and D. Radev, "QMSum: A New Benchmark for Query-based Multi-domain Meeting Summarization," in *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, 2021.
- [14] Gemini Team, "Gemini: A Family of Highly Capable Multimodal Models," arXiv:2312.11805, 2023.
- [15] Google, "Gemini API: Audio Understanding," Google AI for Developers, technical documentation, accessed September 2026.
- [16] Google, "Gemini API: Video Understanding," Google AI for Developers, technical documentation, accessed September 2026.
- [17] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, vol. 1, 2019, pp. 4171–4186, doi: 10.18653/v1/N19-1423.